Zheyuan "Brian" Zhang

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Research Interests

My current research centers on **Embodied AI** and **Continual Learning**, particularly latent world models, test-time training, and persistent memory.

Areas: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Cognitive Science, Developmental Psychology

Education

Johns Hopkins University, Baltimore, MD, US — 08/2025 - 05/2029 (expected)
Ph.D. in Computer Science
Advisor: Tianmin Shu, Daniel Khashabi

University of Michigan, Ann Arbor, MI, US — 08/2022 - 04/2024
M.S. in Robotics
Certificate, Cognitive Science
GPA: 4.00/4.00
Advisor: Joyce Chai

University of Massachusetts Amherst, Amherst, MA, US — 09/2018 - 05/2022
B.S. in Computer Science
GPA: 3.85/4.00

Honors and Awards

Amazon AI PhD Fellow. 2026-2027

Gold Reviewer. International Conference on Machine Learning (ICML). 2026

Oral Presentation (1.8%). International Conference on Learning Representations (ICLR). 2025

Chancellor's Award, $40,000 USD (total). UMass Amherst. 2018-2022

Dean's List Honors (all semesters). UMass Amherst. 2018-2022

Second Prize Excellent Paper Award. INSAI 2021 Conference. 2021

Publications

* → equal contribution. † → equal advising.

[P.1] Yifan Yin*, Zehao Wen*, Suyu Ye, Jieneng Chen, Zehan Zheng, Nanru Dai, Haojun Shi, Aydan Huang, **Zheyuan Zhang**, Alan Yuille, Jianwen Xie, Ayush Tewari, Tianmin Shu. "3D-Belief: Embodied Belief Inference via Generative 3D World Modeling". In *arXiv*. 2026.

[W.1] **Zheyuan Zhang**, Suyu Ye, Nakul Agarwal, Hossein Nourkhiz Mahjoub, Ehsan Moradi Pari, Daniel Khashabi, Tianmin Shu, Vaishnav Tadiparthi. "JEPA-TTT: Persistent Test-Time Training of Latent World Models for Planning under Dynamics Shifts". In *WM PAI Workshop @ NeurIPS*. 2026.

[**W.2**] Suyu Ye, **Zheyuan Zhang**, Vaishnav Tadiparthi, Hossein Nourkhiz Mahjoub, Ehsan Moradi Pari, Tianmin Shu, Homanga Bharadhwaj, Nakul Agarwal. "Watch, Infer, Coordinate: Inferring Robot Partner Constraints for Zero-Shot Coordination". In *PhysUnderstand Workshop @ NeurIPS*. 2026.

[**W.3**] Alvin Zhang*, Zehao Wen*, **Zheyuan Zhang**, Tianmin Shu†, Daniel Khashabi†. "MemAnchor: A Parametric Memory Harness for Context Reconstruction". In *LCFM Workshop @ NeurIPS*. 2026.

[**W.4**] **Zheyuan Zhang***, Alvin Zhang*, Daniel Khashabi†, Tianmin Shu†. "Continual Learning Mechanisms Compose for Long-Horizon Memorization". In *CL4FMAgents Workshop @ NeurIPS*. 2026.

[**W.5**] Hongxin Zhang*, **Zheyuan Zhang***, Zeyuan Wang*, Zunzhe Zhang, Lixing Fang, Qinhong Zhou, Chuang Gan. "Ella: Embodied Lifelong Learning Agents with Non-Parametric Memory". In *MAS Workshop @ ICML*. 2026.

[**C.1**] **Zheyuan Zhang***, Zehao Wen*, Alvin Zhang, Andrew Wang, Jianwen Xie, Daniel Khashabi†, Tianmin Shu†. "AgentOdyssey: Open-Ended Long-Horizon Text Game Generation for Test-Time Continual Learning Agents". In *Advances in Neural Information Processing Systems 40 (NeurIPS)*. 2026.

[**C.2**] Qinhong Zhou*, Hongxin Zhang*, Xiangye Lin*, **Zheyuan Zhang***, Yutian Chen, Wenjun Liu, Zunzhe Zhang, Sunli Chen, Lixing Fang, Qiushi Lyu, Xinyu Sun, Jincheng Yang, Zeyuan Wang, Bao Chi Dang, Zhehuan Chen, Daksha Ladia, Jiageng Liu, Chuang Gan. "Virtual Community: An Open World for Humans, Robots, and Society". In *The Fourteenth International Conference on Learning Representations (ICLR)*. 2026.

[**C.3**] **Zheyuan Zhang***, Fengyuan Hu*, Jayjun Lee*, Freda Shi, Parisa Kordjamshidi, Joyce Chai, Ziqiao Ma. "Do Vision-Language Models Represent Space and How? Evaluating Spatial Frame of Reference Under Ambiguities". In *The Thirteenth International Conference on Learning Representations (ICLR)*, Oral Presentation (1.8%). 2025.

[**C.4**] Yuncong Yang*, Jiageng Liu*, **Zheyuan Zhang**, Siyuan Zhou, Reuben Tan, Jianwei Yang, Yilun Du, Chuang Gan. "MindJourney: Test-Time Scaling with World Models for Spatial Reasoning". In *Advances in Neural Information Processing Systems 39 (NeurIPS)*. 2025.

[**C.5**] Shane Storks, Itamar Bar-Yossef, Yayuan Li, **Zheyuan Zhang**, Jason J. Corso, Joyce Chai. "Transparent and Coherent Procedural Mistake Detection". In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2025.

[**C.6**] Hongxin Zhang*, Zeyuan Wang*, Qiushi Lyu*, **Zheyuan Zhang**, Sunli Chen, Tianmin Shu, Behzad Dariush, Kwonjoon Lee, Yilun Du, Chuang Gan. "Compositional World Models for Embodied Multi-Agent Cooperation". In *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025.

[**C.7**] **Zheyuan Zhang**. "A Combinatorial Approach to Neural Emergent Communication". In *Proceedings of the 31th International Conference on Computational Linguistics (COLING)*. 2025.

[**C.8**] Keunwoo Peter Yu, **Zheyuan Zhang**, Fengyuan Hu, Shane Storks, Joyce Chai. "Eliciting In-Context Learning in Vision-Language Models for Videos Through Curated Data Distributional Properties". In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2024.

[**C.9**] **Zheyuan Zhang***, Shane Storks*, Fengyuan Hu, Sungryull Sohn, Moontae Lee, Honglak Lee, Joyce Chai. "From Heuristic to Analytic: Cognitively Motivated Strategies for Coherent Physical Commonsense Reasoning". In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2023.

[**C.10**] **Zheyuan Zhang**, Huiliang Shang. "Low-cost Solution for Vision-based Robotic Grasping". In *Proceedings of the 2021 International Conference on Networking Systems of AI (INSAI), Second Prize Excellent Paper Award*. 2021.

Research Experience

Honda Research Institute, Research Intern — San Jose, CA, US
Adviser: Hossein Nourkhiz Mahjoub, Vaishnav Tadiparthi — 05/2026 - 08/2026

University of Massachusetts Amherst, Research Intern — Cambridge, MA, US
MIT-IBM Watson AI Lab, Visiting Researcher — 06/2024 - 05/2025
Adviser: Chuang Gan

University of Michigan SLED Research Lab, Research Assistant — Ann Arbor, MI, US
Adviser: Joyce Chai — 11/2022 - 05/2024

LG AI Research, Research Collaboration — Ann Arbor, MI, US
Adviser: Honglak Lee — 11/2022 - 11/2023

Fudan University RAS Lab, Research Intern — Shanghai, CN
Adviser: Huiliang Shang, Ruijiao Li — 06/2021 - 08/2021

Teaching

Graduate Student Instructor (GSI) — Winter 2024, Fall 2023
EECS 492: Introduction to Artificial Intelligence
Topics: Search, Constraint Satisfaction Problems, Logic and Inference, Uncertainty, Bayesian Networks, Decision Trees, Linear Regression, Neural Networks and Generative AI, Nonparametric Methods, Decision Theory, Reinforcement Learning, Game Theory

Guest Lectures

Apr. 2024	University of Michigan, Host: Alexander Rodríguez, Laura Wendlandt
Nov. 2023	University of Michigan, Host: Emily Mower Provost, Alexander Rodríguez

Academic Service

Workshop Organizing

- Leading Organizer, Towards Test-Time Continual Learning Agents (TTCL @ NeurIPS 2026).

Journal Reviewing

Machine Learning	TMLR

Conference Reviewing

Machine Learning	ICLR, NeurIPS, ICML
Natural Language Processing	ACL

Workshop Reviewing

Natural Language Processing	ACL-SRW, EACL-SRW
Machine Learning	NeurIPS Pluralistic Alignment
Robotics	RSS Continual Robot Learning from Humans

Other Projects

1. Bot Lab: Autonomous Ground Vehicle from Low-level Control, SLAM to Planning and Exploration — 2022
2. Clara in Wonderland: 3D Open-world Adventure Game (Unity, C#) — 2021

3. Quanin: Automatic Stock Screener (Python, C#) 2021

4. Blockchain From Scratch (C++) 2020

5. Findurcourse.com: Node.js Web Application (HTML/CSS, JavaScript, PostgreSQL) 2020

6. ZiZoyaOS: 32-Bit Operating System From Scratch (Assembly, C) 2020

Skills

- Programming Languages: Python, C/C++, C#, JavaScript, MATLAB, Visual Basic, Pascal
- Other Computer Languages and Software: HTML/CSS, SQL, LaTeX, Blender, Unity, Unreal Engine